DBMS Assignment (1)

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Q1) What are the key reasons why a DBMS is preferable to a traditional file system how does it ensure data security and consistency?

~~Q2) Ans 1)~~

Q2) A hospital wants to store patient, doctor and appointment data in DBMS Identifies the entities, attributes and relationship in the system. Draw the ER diagram and explain the key design decisions you made.

Q3) An online shopping platform needs to store data about customers, orders and products. Suggest primary keys foreign keys and the integrity constrains for the system. How do these constrains help maintain the accuracy of the data.

Q4) Why is understanding Entity Relationships models important when designing a database? What could happen if a relationship is not correctly defined between entities

Ans => P.T.O

1) Reasons for why DBMS is preferred:

a) Reduced data redundancy
b) data consistency
c) Data integrity
d) Automatic recovery after crashes
e) It manages multiple users accessing data at the same time.

It ensures data security and consistency by -

- Authentication
- Authorization
- Encryption
- Integrity Constrains (Prevents Invalid data entry)

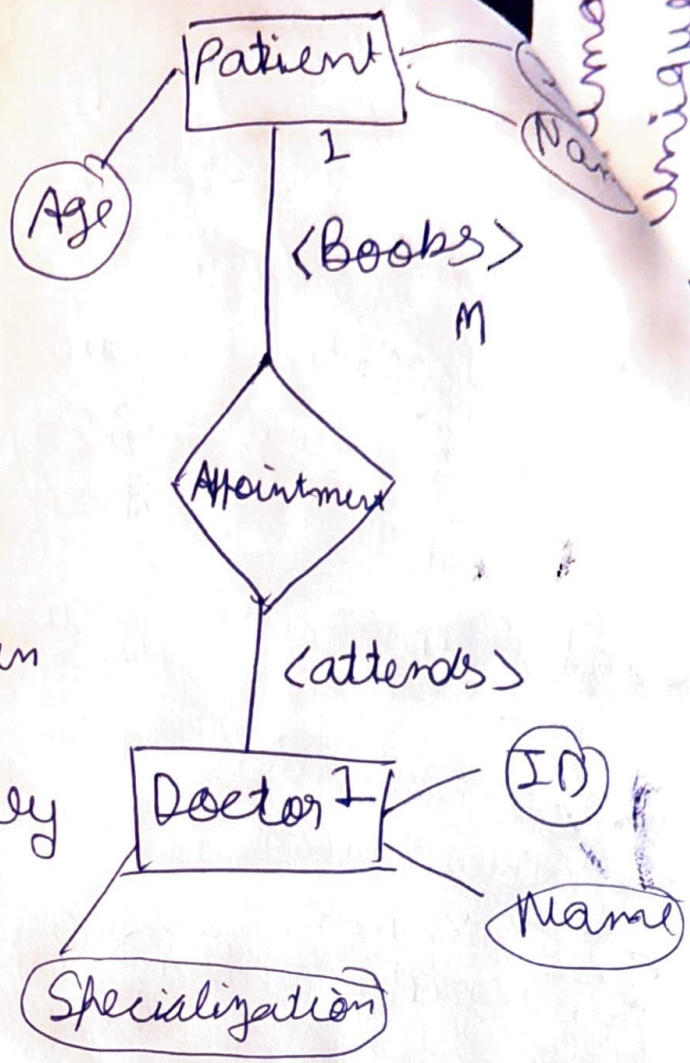
ans2) Entities and Attributes :

- Patient — Patient ID, Name, Age
- ~~Doctor~~
- Doctor — Doctor ID, Name, Specialization
- Appointment — Appointment ID, Date and Time, Patient ID, Doctor ID

Relationships :

- Patient can book many appointments
- Doctor can have many appointments.
- Appointments connect doctors and Patients

~~Patient~~
~~1 <Books> M~~
~~Appointment~~
~~M <attends> 1~~
~~Doctor~~




Key Decisions:
1) Primary keys uniquely identify records
2) Foreign keys maintain referencial Integrity
3) Avoided Reduadancy by separating entities

ns 3) Entities:

1) Customer
- ID
- Name
- Phone

2) Product
- ~~Product~~ ID
- ~~Product~~ name
- Price

3) Order
- Order_ID
- Order_Date
- Customer ID

4) Order details
- Order_ID
- Product_ID
- Quantity

nary key:
uquely identifies each record

Foreign key:
Maintains relationship between tables
Example: Customer_ID in order
Product_ID in order details

Integrity constrains
→ Entity Integrity
→ Referall Integrity
→ Domain const.
- Price ≥ 0
- Stock ≥ 0
→ check constrains
Quantity > 0
Constrains maintains accuracy by:
1) Prevent duplicate entries
2) Prevent invalid references
3) Prevent negative price and stock
4) Ensure Invalid data

ans 4) → E R models are important because:
- helps in visualization of data structure
- Improve data base efficiency
- Clearly defines entities and relation-ships.
- Prevent Redudancy
- Makes development easier

What happens if relationships are not correctly defined:

1) data inconsistancy
2) Integrity issues
3) update anomalies
4) Deletional anomalies
5) Poor query performance.

Example:

If order and product relationship is not properly defined:

→ Some products may be repeated multiple times.

→ Deleting orders might delete product details accidently.